

Joint ESA/DLR Call : Join the Science Definition Team of the COMPACT Plasma Experiment

Contribute to the next dusty plasma experiment in microgravity !



Director of Human and Robotic Exploration - Daniel Neuenschwander

Submit Your Idea

German Space Agency

Within the "National space programme for innovation and international cooperation", the German Space Agency at DLR provides national funding for space research and the development of innovative technologies. This not only drives exploration and scientific knowledge gain, but also provides key contributions to societal challenges like health, climate protection, mobility, communication, or safety.

One of the programme's focus areas, "Research and Exploration", covers biological, human physiological, physical, materials science, and exploration topics. It funds scientific research with a space background and/or realises experiments which utilise specific space conditions for their research objective – like for instance absence of gravity. To provide or simulate space conditions, access to Earth-based space analogues and different microgravity platforms like drop towers, sounding rockets, the International Space Station (ISS), and parabolic flights is offered by the German Space Agency at DLR. Through contracts with the space industry, multi-user facilities are provided with which overarching topics in microgravity research can be investigated. One such facility currently in development is the Complex Plasma Facility (COMPACT).

This call for ideas falls within the domain of plasma physics, and in particular addresses the topic of dusty or complex plasmas under microgravity conditions. ESA's role is to facilitate access to the wider European scientific community and to support the identification of scientific and technical inputs that may strengthen the case for COMPACT as a future multi-user facility for low-Earth orbit research.

Complex plasmas and the COMPACT facility

Complex or dusty plasma is a state of soft matter that consists of micrometer-sized particles immersed in a low-temperature plasma. The particles become charged by plasma constituents (electrons, ions) and interact with each other and their surroundings via the electric force. Depending on the particle charge, density and temperature, the interaction can become very strong, and leads to collective effects and the occurrence of solid or liquid properties. Complex plasmas are a unique physical system that allow to study fundamental many-particle physics on the kinetic level of the individual particles. For these investigations, a microgravity environment is essential since the gravitational force causes a sedimentation of particles that can prevent the formation of three-dimensional clouds of particles.

The laboratory complex plasma is a state of soft matter where known microparticles are immersed in a weakly ionized, low-density gas – a low-temperature plasma. The particles acquire a charge in the plasma that scales with the surface potential. This provides a strong Coulomb interaction and therefore strong coupling between the microparticles and allows studying gaseous, liquid and crystalline states of the particle arrangements as well as transitions between them. The collection of particles is optically thin, so that imaging can be used to track ensembles of micron-size particles, measuring their individual coordinates and velocities. This is a unique feature of a complex plasma. With currently available optical diagnostics, all time scales relevant for the system dynamics can be easily resolved. Therefore, complex plasmas can be regarded as model systems for classical condensed matter physics where phenomena can be observed through dynamical tracking of individual particles (“proxy atoms”) in a virtually undamped system. Complex plasmas build a bridge between single particle dynamics over mesoscopic systems to fully statistical systems.

Gravity strongly affects the behavior of complex plasmas due to the high mass of the microparticles as illustrated in Fig. 1. Within the main plasma volume, there are no forces such as buoyancy that can offset gravity, and the virtually undamped system of microparticles sediments into 2D, quasi-2D and stressed 3D systems close to the lower plasma boundary (the plasma sheath), where electric forces levitate the particles against gravity. Those systems can in general be used for a number of fundamental investigations such as particle charging and interaction, waves, phase transitions and crystallization dynamics in 2D and stressed 3D systems, or hydrodynamic instabilities, shear flow, heat transport, anisotropic interactions in 2D systems and more. Planetary physics questions focusing on natural dusty plasmas, e.g. on the Moon, can also be addressed such as: what is the charge of lunar dust regolith in the local space plasma environment, are the local electric fields building up on the surface strong enough to transport, loft or even levitate dust, are the dust particles attracted by the surfaces of space crafts or space suits?

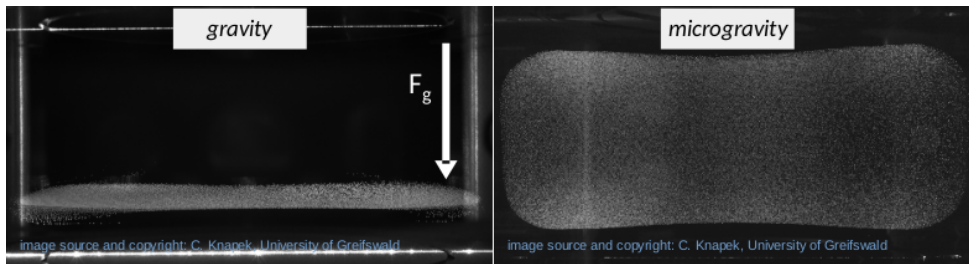


Fig. 1 : Complex (dusty) plasmas under gravity (two-dimensional "flat plane") and in microgravity (three-dimensional volume)

COMPACT will be the continuation of the successful complex plasma research on the ISS (PKE-Nefedov, PK-3 Plus and the currently operated PK-4). It is intended to be a multi-purpose, multi-user facility with international accessibility and the involvement of a larger science community. A technical article on the COMPACT facility has been published in open access in 2022. You can access it following [this link](#) .

The former PKE-Nefedov and PK-3 Plus facilities were based on radio-frequency discharges optimized for the generation and study of 3D, stable microparticle suspensions. Fundamental properties of complex plasma such as particle charging or wave propagation were investigated as well as more complex phenomena such as crystallization and melting processes, electrorheological effects in complex plasmas, or lane formation and phase separation in binary mixtures of microparticles. The recent PK-4 facility is based on a dc discharge tube designed to study fluid and flowing complex plasmas. Aside basic dust-plasmas related phenomena (e.g. dust ionization waves or dust-induced striations), structural (string-like order) and transport (shear viscosity and heat diffusivity) properties of the dust component were investigated.

The COMPACT facility will again be based on a radio-frequency discharge. Compared to the previous facilities, the discharge volume is 10 times larger and, hence, will allow the generation of much larger and more homogeneous 3D particles systems. It offers a multitude of particle and plasma manipulation options that were not available on the previous facilities (compression/expansion of a particle cloud by movable electrodes, four independent RF electrodes, pulsed plasma mode, superposition of DC/slow AC bias on the electrodes, infrared laser for local particle manipulation, UV-lamp for charge manipulation). Further, advanced optical diagnostics offer unique opportunities: the conventional 2D particle diagnostics is complemented by a stereoscopic camera system yielding real-time 3D particle information, and plasma glow is observed by a hyperspectral camera.

The experimental ideas that were the major design drivers for COMPACT come from the areas of statistical physics,

phase transitions, nonlinear dynamics, active matter, planetary dust and dust-plasma interactions. Some prominent examples are studies on the equation of state of a complex plasma, the elementary mechanisms which determine the stability of supercooled liquids against crystallization, or how microscopic interactions lead to the development of large-scale nonlinear (turbulent) motion. Introducing active particles into the plasma opens a new research direction covering many aspects of active matter such as the structure and ordering, or how the collective behaviour of self-propelled particles connects to the energy input on the single-particle level. In addition to active particles, non-spherical particles (such as ellipsoids or dumbbells) might also be used. The planetary physics questions focus on natural dusty plasmas, e.g. on the Moon, with the aim to shed light on questions such as charging of lunar dust regolith in the local space plasma environment, and lofting of dust. Additionally, methods can be developed to utilize particles in a plasma as probes to study the plasma environment itself without the need of further diagnostics, which is beneficial in a remote space experiment with limited resources.

More information can be found on the [COMPACT website](#).

References:

- C. A. Knapek, L. Couëdel, A. Dove, J. Goree, U. Konopka, A. Melzer, S. Ratynskaia, M. H. Thoma, H. M. Thomas ; [COMPACT - a new complex plasma facility for the ISS](#) ; *Plasma Phys. Control. Fusion* **64** 124006 (2022)
- M. Pustynnik, H. M. Thomas, M. Kretschmer, M. Thoma ; [Plasmakristall-4 Experiment: 10 Years of Operation in Orbit](#) ; *Contributions to Plasma Physics*, 0:e70063 (2025)
- H. M. Thomas, M. Schwabe, M. Y. Pustynnik, C. A. Knapek, V. I. Molotkov, A. M. Lipaev, O. F. Petrov, V. E. Fortov, S. A. Khrapak ; [Complex plasma research on the International Space Station](#) ; *Plasma Phys. Control. Fusion* **61** 014004 (2019)

Objective of This Call

The main objective of this call is to identify further science cases (science extensions) in fundamental research, planetary sciences and exploration beyond the existing COMPACT Science Definition Team (SDT) to improve the scientific case for flying COMPACT by engaging the wider ESA and DLR scientific communities.

Another objective of this call is to find new or alternative approaches to reduce the footprint of COMPACT while maintaining its scientific content (hardware simplifications), so that it fits more low-earth orbit (LEO) destinations and more easily secures a flight opportunity in the future.

The science teams behind selected ideas will join the existing scientific team of COMPACT.

Calendar of the call

- TBD October 2026 : information session for Science Teams
- 30th November 2026 : Submission Deadline
- NET 31st December 2026 : Decision notification to Science Teams

SPECIAL CONDITIONS

Coordinators proposing ideas must be affiliated or jointly affiliated with a scientific institution, including universities, public research institutes (e.g. *CNRS*, *Max-Planck Gesellschaft*, *Wigner Institute* ...) or a public administration (e.g. *DLR*).

Collaboration with existing research teams already working on COMPACT is appreciated ; external scientists are also encouraged to suggest new ideas.

The scientific institution for which the coordinator of an idea is working must be located in one of the ESA member states: Austria, Belgium, Canada, Czech Republic, Denmark, France, Estonia, Finland, Germany, Hungary, Ireland, Italy, Luxembourg, the Netherlands, Norway, Poland, Portugal, Romania, Slovenia, Spain, Sweden, Switzerland and the United Kingdom.

Scientific teams interested in submitting an idea are advised to contact their national space agency or national delegation to clarify possible support mechanisms applicable in their country. Any financial or institutional support to participating scientific teams, where available, is expected to be handled through the relevant national agency. A list of national points of contact is available [following this link](#).

EVALUATION CRITERIA

The scientific content of all ideas will be evaluated, including by the existing science team of COMPACT. The evaluation criteria will slightly differ whether the received ideas consist of *science extensions* (additional experiments), *hardware*

simplifications, or both.

1. **Scientific Impact (40 %)** : For science extensions, the scientific impact will measure how much additional science can be made by implementing the requested changes, and how impactful these experiments will be, notably in comparison with the previous microgravity plasma experiments. For hardware simplifications, we will assess how the requested changes will adversely (or positively) affect the original scientific programme of COMPACT.
2. **Impact on Payload Design (40 %)** : This criterion encompasses the ease of implementation of the requested changes ; the impact (negative or positive) on hardware mass, volume and power consumption ; and the readiness levels of the proposed technologies.
3. **Team quality and experience (20%)** : We will positively receive ideas from teams encouraging profile diversity (senior / early career researchers) ; the overall experience of the team in the topic of (experimental) complex plasmas and microgravity research will also be assessed.

Only ideas scoring **81** and higher (on average) and **71** (for each criterion) will be eligible for implementation. The final decision on implementation will be given based on additional discussions between the existing science team of COMPACT and the German Space Agency (DLR).

ATTACHMENTS

Name	Added on
 ESA Privacy Notice for HRE-S Activities.pdf	Jul 7
 Science Team Members.xlsx	Jul 7
 Points_of_Contacts_National_Agencies_Update_2026.pdf	Jul 7
 OSIP-General_Conditions_of_Participation_v4.pdf	Jul 31

Do you need help?



Campaign Manager - Brice Saint-Michel, Project Scientist

Background Information

INNOVATION AREA

Human and Robotic Exploration

ESA's [Terrae Novae exploration programme](#) is leading Europe's human journey into the Solar System using robots as precursors and scouts. Exploring space is about travelling farther and coming back with new experiences and knowledge to help us on Earth. Humankind will benefit from the new discoveries, ambitions, science, inspiration and challenges.

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